

Understanding the Analysis Notebook

Phase 5 explained, one question at a time

What is this phase for?

In the previous phase (EDA) we got to know the data. We were looking around, checking what was there, spotting anything that might trip us up.

This phase is different. Here we **answer the actual business questions** — the ones the project was set up to answer. Which habitat needs protecting? Where should birdwatchers go? Which birds depend on which habitat?

The important word is **guarded**. Every answer here obeys four rules we worked out earlier. Those rules exist because this data has traps in it, and without the rules you get answers that look convincing and are wrong.

The four guardrails, in plain words

Rule	What it means	Why
G1	Never compare raw counts between habitats. Always compare rates per survey.	One habitat was surveyed far more than the other. Raw counts measure how much looking happened.
G2	Only compare habitats inside the same park, and only in the four parks that have both.	Seven parks have no grassland at all. Including them compares different places, not different habitats.
G3	Measure per survey session, never per plot.	Grassland plots were revisited more often than forest plots.
G4	Charts show adjusted rates, never raw totals.	A misleading chart is more convincing than a misleading table.

The headline: three problems we caught

Before the individual questions, it is worth knowing that this phase found **three separate errors** that would otherwise have gone into the final report. None of them were maths mistakes. All three came from asking "is this number really saying what it looks like it is saying?"

What we caught	What would have happened
One bird recorded under two different scientific names	A single ordinary bird would have appeared as two different "specialist" species, one for each habitat.
82% of our headline finding came from one species	We would have claimed a general pattern across at-risk birds, when it is really about one bird.
A "strong result" that was just more surveying	We would have reported grassland as having five times more exclusive species. At equal effort it is roughly even.

Cell 1 — Setting up the rules

The question

Load the data, and make the four guardrails explicit in the code.

What the code did

It opened the same four tables as before, then created one important thing: a filter called **shared**, which keeps only the survey sessions from the four parks that have both forest and grassland.

From this point on, any habitat comparison uses *shared* rather than the full dataset. The rule is built into a variable rather than left as something to remember.

What we found

Of 1,408 total sessions, **737** are in the four shared parks. So slightly more than half the data is usable for comparing habitats.

Why it matters

Writing the rule into the code, rather than trusting yourself to remember it, is the difference between a rule and a good intention. Later, when you are tired and building the tenth chart, the filter is already there.

Q1 — Which habitat holds more at-risk birds?

The question

Where should conservation effort go?

What the code did

For each of the four shared parks, it worked out what percentage of bird sightings were species on the official at-risk list — separately for forest and grassland. Then it checked whether forest came out higher in *every* park, or just on average.

What we found

Forest **3.65%** against grassland **0.59%** — about **6.2 times** higher. And forest was higher in all four parks, with no exceptions.

An everyday way to think about it. If you tested four schools and found the same result in all four, you would trust it more than if the overall average looked good but two schools disagreed.

Why it matters

The "all four parks" check is the important part. An average can be produced by one unusual place. If the pattern holds everywhere you look, it is much more likely to be real.

The statistical test came back with a p-value of about 0.000000000000000002. That means: if there were genuinely no difference, you would have to repeat this study an unimaginable number of times to see a gap this large by luck.

Q2 — Is one habitat more biodiverse?

The question	Do you find more different kinds of bird in forest or in grassland?
--------------	---

What the code did

It calculated the answer **twice, deliberately**. First the tempting way — lumping all 11 parks together. Then the correct way — only inside the four parks that have both habitats.

What we found

The two answers disagree completely.

All 11 parks lumped together: grassland **9.15** species per survey against forest **8.61**, and the test says this is a real difference.

Inside the four shared parks: grassland **9.15** against forest **9.10**. No difference at all.

An everyday way to think about it. Imagine comparing two doctors' success rates. One only treats mild cases, the other takes on the hardest ones. Lump the numbers together and the first looks better. Compare them on similar patients and the difference disappears.

Why it matters

Here is what happened. Forest was surveyed in seven extra parks that grassland was never surveyed in. Those seven parks happen to have fewer species. So they dragged the forest average down — and grassland never suffered the same drag, because it was never there.

The first comparison was not measuring habitat. It was measuring **which parks were in which group**.

This has a name: **Simpson's paradox**. A pattern that appears when you lump everything together, and vanishes or reverses when you split it properly.

How to report a result like this

"No difference" feels like failure. It is not — it is a finding, and here it is a more interesting one than a positive result would have been.

Weak version	"We found no significant difference in species richness between habitats."
Strong version	"A raw comparison suggests grassland is significantly richer (9.15 vs 8.61). However, grassland was surveyed in only 4 of 11 parks. Restricting the comparison to parks with both habitats removes the difference entirely (9.15 vs 9.10), and the direction is inconsistent between those four parks. The apparent advantage reflects which parks were surveyed, not the habitat."

The second version shows you found a trap and stepped around it. That is worth more than any positive finding.

Q3 — Where should birdwatchers go?

The question

Which parks and plots give you the most birds for your morning?

What the code did

It ranked all 11 parks by average species per survey session — then ranked them again by raw species count, to show the two answers differ.

What we found

By species per visit: **ANTI** comes first (10.04), then CHOH, NACE, WOTR.

By raw species count: **MONO** comes first with 100 species — but MONO drops to sixth once you account for effort, because it was surveyed 232 times.

An everyday way to think about it. A restaurant with one five-star review is not better than one with four hundred reviews averaging 4.6. The average alone does not tell you how much to trust it.

Why it matters

For someone deciding where to spend one morning, ANTI is the better answer: you see more per trip. For a conservation inventory, MONO's 100 species matters more. Different questions, different rankings — and the raw list gives the wrong answer to the tourism question.

But watch out for a new problem. WOTR ranks fourth on only **12 sessions**. ANTI's first place rests on 278. Twelve mornings is not enough to trust a rate. Normalising for effort fixed one bias and introduced sensitivity to small samples.

Q4 — Which birds depend on which habitat?

The question

Are there species that would lose their home if one habitat disappeared?

What the code did

For every species with at least 20 sightings in the shared parks, it worked out what share of those sightings were in grassland. Over 90% means grassland specialist, under 10% means forest specialist, anything between means generalist.

What we found

17 grassland specialists. Zero forest specialists. 31 generalists.

But the first time we ran this it said 18 and 1 — and the "1 forest specialist" turned out to be a bug.

Why it matters

The asymmetry is the finding: grassland supports a group of birds found almost nowhere else, while the forest birds here are all adaptable enough to use both. That is a direct argument for protecting grassland.

The bug we found here

The same bird — American Goldfinch — appeared in **both** the grassland specialist list and the forest specialist list, with contradictory numbers. That is impossible, so something upstream was wrong.

It turned out the species was recorded under two different scientific names. **Carduelis tristis** in the grassland file, **Spinus tristis** in the forest file. Same bird — the four-letter birding code is AMGO for both. The species was reclassified into a different genus at some point, and the two spreadsheets were made using different vintages of the naming system.

Because the grassland file used one name and the forest file used the other, the split fell **exactly** along habitat lines: 310 grassland sightings under one name, 147 forest sightings under the other. Neither half had any sightings in the other habitat, so both looked like perfect specialists.

The giveaway

A real specialist has a lopsided split, like 5 forest and 346 grassland. A perfectly clean 0-and-310 split is suspicious. Nature is rarely that tidy.

The fix uses the four-letter code as the authority, because that code does not change when scientists rename a genus. After fixing it, American Goldfinch is one bird, 89.1% grassland, correctly classified as a generalist — and there are **zero** forest specialists, which makes the asymmetry finding stronger, not weaker.

Q5 — Which at-risk birds, and where?

The question

The 6.2x finding sounds impressive. What is actually behind it?

What the code did

It listed every at-risk species with its forest and grassland counts, and how many parks it appears in.

What we found

There are only **8** at-risk species in the whole dataset. And one of them — **Wood Thrush** — accounts for **309 of the 378** at-risk sightings. That is **82% of the entire finding from a single bird**.

An everyday way to think about it. If a shop reports record sales, and 82% came from one customer, that is a different business story than "sales are booming". Both are true. Only one is useful.

Why it matters

This changes what we can honestly claim.

We re-ran the comparison with Wood Thrush removed. The gap shrank from **6.2 times to 2.1 times**, and the "holds in all four parks" claim dropped to holding in only two of four.

So the honest finding is narrower than it first appeared. It is not "at-risk birds generally prefer forest". It is "**one widespread, forest-dependent at-risk species dominates the region's at-risk records**".

This is arguably a **better** finding. "Protect forest for at-risk birds generally" is vague. "Wood Thrush is forest-dependent, present in 10 of 11 parks, and is the dominant at-risk species here" is something a land manager can act on this week.

Q6 and Q7 — When are birds most active?

The question

Does it matter what month or what time of morning you survey?

What the code did

It compared species per session across months, then across time-of-day bands, then directly tested whether starting early beats starting late.

What we found

Grassland: 9.70 species if you start at 5–6am, 8.25 if you start at 9–10am. That is 1.46 more species per visit, and the test says it is solidly real.

Forest: 8.65 early versus 8.36 late. A difference of 0.30, which the test says is indistinguishable from random noise.

An everyday way to think about it. Across 100 grassland surveys, going early rather than late gains you roughly 146 extra species detections, for no extra cost at all.

Why it matters

This is the most directly usable finding in the whole project. Same observer, same ten minutes, same plot — just an earlier start — and you record **18% more species** in grassland.

And because forest shows no such effect, the recommendation is specific rather than blanket: **put your early-morning slots into grassland; schedule forest whenever suits.** That tells a manager where to spend their limited early capacity.

Q8 and Q9 — Does the weather matter?

The question

Do temperature, sky and disturbance change what gets recorded?

What the code did

It grouped surveys into weather bands and compared the averages, then tested whether the worst disturbance level really differs from none.

What we found

Temperature matters and the shape is interesting: richness peaks at 15–20°C (9.61) and falls off *both* ways, down to 7.97 above 30°C. It is an optimum, not a straight line.

Partly cloudy beats clear sky (9.20 versus 8.75). Bright sun is not ideal birding weather.

Serious disturbance is the strongest effect of all: 5.65 species against 8.93 when undisturbed. That is a **37% loss**.

Why it matters

These translate into practical guidance: survey in mild, partly cloudy conditions, and always record disturbance because it costs you more than a third of what you would otherwise find.

Two things we deliberately did not smooth over

An anomaly we cannot explain. "Slight disturbance" scored *higher* (9.38) than "no disturbance" (8.93). That should not happen, and both categories have hundreds of sessions so it is not a small-sample fluke. We do not know why. The right response is to report it as observed and not invent a story. A report that admits an unexplained result is more trustworthy than one that quietly tidies it away.

A number that looked impressive and was not. The driest humidity band showed 11.50 species — the highest figure in the table. It came from **12 sessions**. Compare that with 8.68 from 704 sessions in the middle band. Remove the twelve and humidity is essentially flat. This is the same trap as WOTR's ranking: once you switch to rates, small samples produce dramatic-looking numbers that mean nothing.

Q10 to Q12 — Checking our own work

The question

Are there problems with how the data was collected that we should disclose?

What the code did

Three checks: do the three observers record differently, how many species are exclusive to one habitat, and how much of the data is actually usable for comparison.

What we found

Observers differ by 37% — but were spread almost perfectly evenly across habitats, so their differences cancel out.

Coverage: only 4 of 11 parks have both habitats, so only 737 of 1,408 sessions can be used for habitat comparison.

Exclusive species looked dramatic: 8 forest-only against 37 grassland-only in the shared parks. Nearly five times as many.

Why it matters

That last number turned out to be our **third** trap.

The third trap: counting things is effort-sensitive too

In the shared parks, grassland had **594 survey sessions and forest only 143** — more than four times as many chances to stumble across a rare bird. And 41% of those "grassland-only" species were seen exactly **once**.

So we tested it properly. We randomly cut grassland down to 143 sessions — matching forest — and recounted, two hundred times over. At equal effort the answer is roughly **21 grassland-only against 17 forest-only**. Close to even. The dramatic gap was almost entirely extra surveying.

The technique

Cutting the bigger sample down to match the smaller one, so counts are comparable, is called **rarefaction**. It is standard practice in ecology and a good word to use in your methodology section.

What makes this worth understanding: we already had two rules protecting us from survey-effort problems. G1 fixed rate comparisons. G2 fixed park composition. Neither of them reaches this. *Counting how many different species you found* stays effort-sensitive no matter how carefully you normalise your rates, because rare species simply keep turning up the longer you look.

The specialist finding from Q4 survives untouched, and here is the lucky reason: it only classifies species with at least 20 sightings. That threshold quietly excludes exactly the rare one-off species that effort inflates. It was doing protective work we had not designed it to do.

What this phase produced

Question	Answer	Confidence
Which habitat for conservation?	Forest — but specifically for Wood Thrush, which is 82% of the at-risk records	High, correctly framed
Is one habitat more biodiverse?	No difference once effort and park are accounted for	High — a null result
Where should birdwatchers go?	ANTI, CHOH, NACE by species per visit	Good, with a caution on small parks
Which birds depend on habitat?	17 grassland specialists, zero forest specialists	High
When to survey?	Grassland before 8am gains 1.46 species per visit; forest timing does not matter	High
What conditions?	Mild and partly cloudy; serious disturbance costs 37%	High
How many exclusive species?	Roughly even at equal effort (21 vs 17)	Moderate — needs rarefaction
Any collection problems?	Observer variation of 37%, but balanced so harmless	Verified

The thing worth taking away

Every one of the three errors we caught in this phase had the same shape. A number looked impressive. Nobody had asked what was really producing it.

The goldfinch split looked like two specialists. The 6.2x looked like a general pattern. The 8-versus-37 looked like a strong asymmetry. All three were real numbers, correctly calculated, and all three would have been misleading in a report.

The habit that caught them is simple and worth keeping: **when a result looks strong, ask what would have to be true for it to be an artefact — and then go and check.** It costs a few minutes. It is the difference between an analysis that survives scrutiny and one that only survives until someone looks closely.